29. $\Gamma\left(\frac{3}{2}\right)$

(a) $\sqrt{\pi}$ (b) $\frac{\sqrt{\pi}}{2}$ (c) $2\sqrt{\pi}$ (d) $\frac{3}{2}$

30. $\int_0^\infty x^4 e^{-x} dx$ is

(a) 12 (b) 4 (c) 4! (d) 64

Miscellaneous problems

Evaluate the following integrals:

1. $\int \frac{1}{\sqrt{x+2}-\sqrt{x+3}} dx$
2. $\int \frac{dx}{2-3x-2x^2}$
3. $\int \frac{dx}{e^x+6+5e^{-x}}$
4. $\int \sqrt{2x^2-3}\ dx$
5. $\int \sqrt{9x^2+12x+3}\ dx$
6. $\int (x+1)^2 \log x\, dx$
7. $\int \log\left(x-\sqrt{x^2-1}\right) dx$
8. $\int_0^1 \sqrt{x(x-1)}\, dx$
9. $\int_{-1}^1 x^2 e^{-2x} dx$
10. $\int_0^3 \frac{x\, dx}{\sqrt{x+1}+\sqrt{5x+1}}$

Summary

In this chapter, we have acquired the knowledge of

- **The relation between the Primitive function and the derived function:**

  A function $F(x)$ is said to be a primitive function of the derived function $f(x)$, if $\frac{d}{dx}[F(x)] = f(x)$

- **Integration of a function:**

  The process of determining an integral of a given function is defined as integration of a function

- **Properties of indefinite integrals:**

  $\int a f(x)\ dx = a \int f(x)\ dx$

  $\int [f(x) \pm g(x)]\, dx = \int f(x)\ dx \pm \int g(x)\ dx$

- **Standard results of indefinite integrals:**

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + c$
2. $\int \frac{1}{x} dx = \log|x| + c$
3. $\int e^x dx = e^x + c$
4. $\int a^x\ dx = \frac{1}{\log a} a^x + c, a > 0$ and $a \neq 1$
5. $\int \sin x\, dx = -\cos x + c$
6. $\int \cos x\, dx = \sin x + c$
7. $\int \sec^2 x\, dx = \tan x + c$
8. $\int \operatorname{cosec}^2 x\, dx = -\cot x + c$
9. $\int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + c, n \neq -1$
10. $\int \frac{f'(x)}{f(x)} dx = \log|f(x)| + c$
11. $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)} + c$

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